Student proposed?	*N*	*If Y, student name*
ID:	**YM-07**	
SUPERVISOR:	Dr Yaaseen Martin	
TITLE:	Developing a Course-specific Chatbot for Student Support	
DESCRIPTION:	This project focuses on the development of a custom chatbot for one of the signal processing courses at UCT. These are often cited as being among the more challenging courses offered in the Electrical Engineering department (as evidenced by some of the higher failure rates), and students often require additional support on top of the usual tutorial sessions and the hands-on examples covered in class.  This project thus aims to provide a support tool to struggling students, where the custom chatbot could be used to assist with understanding core concepts of the field, identifying errors in calculations, showing mathematical workings and proofs, answering queries about the course, and more. Appropriate benchmarks will need to be discussed as part of the student's final report, and a variety of use cases will need to be tested; students will also need to train the chatbot on existing course notes, examples, tests and so on. Ideally, the chatbot should also be easily adaptable to new courses.	
DELIVERABLES:	A report detailing the relevant literature and background theory, as well as a proposed design to meet the criteria and a working implementation of the chatbot with sufficient accuracy and reliability.	
SKILLS/REQUIREMENTS:	EEE4114F is **required**. Strong performance in EEE2047S, EEE3092F, and/or EEE4114F is also recommended as the student should be well versed in the relevant signal processing content.	
GA 1: Problem solving: *Identify, formulate, analyse and solve complex* engineering problems creatively and innovatively*	*This project involves designing and developing a machine learning-based chatbot system with capabilities for assisting students with material from one of the signal processing courses offered at UCT. Part of the problem-solving that the student needs to engage in is determining the operational requirements of the model, such as performance and reliability, and selecting appropriate training data and testing methods. The selection of suitable tools and implementations will thus need to be clearly justified in the report.*	
GA 4: Investigations,**	*The performance and reliability of the developed model(s) will need to be analysed. The student will need to investigate and justify their design choices and perform appropriate*	

experiments and analysis: *Demonstrate competence to design and conduct investigations and experiments.*	*analyses to determine the chosen model's suitability for its intended purpose. It is important that all design criteria and experimental procedures are also documented to assist users that may want to further develop or adapt the chatbot app in the future.*
GA 5: Use of engineering tools: *Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems*	*The aim of this project is to develop a course-specific chatbot for assisting students with signal processing concepts and practice. The performance, limitations and reliability of these models will thus need to be analysed, and the student will need to make use of various development and programming tools (such as an IDE) as part of their software implementation and design phases. Benchmarks and verification tests will also need to be conducted to determine the developed model's performance, and justifications will need to be made for each of their tools and design choices. Results will also need to be formed by running and modifying code, and by then developing figures, plots, and tables (among other representations) — all of which will require extensive use of tools for the purposes of data analysis, comparisons between solutions, benchmarking, and more.*
EXTRA INFORMATION:	https://ieeexplore.ieee.org/abstract/document/10493063 https://peer.asee.org/creating-and-implementing-a-custom-chatbot-in-engineering-education
BROAD Research Area:	Machine learning, Engineering education
Project suitable for ME/ ECE/EE/ALL?	ALL

***NOTE: Complex engineering problems** require in-depth fundamental and specialized engineering knowledge and have one or more of the characteristics:

- are ill-posed, under- or overspecified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

****NOTE: GA 4:** The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline. An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artifact could be produced.

Ethics clearance questionnaire

		Yes	No
Q1	Does this project involve data collection		X
Q2	Does this project involve utilizing a third-party data set	X	
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?	X	
Q4	Does it exceed the minimum risk defined here: Link [Answer is No here if your project does not utilize ML and AI]		X

Q5	Does this project involve external parties, funders, etc		X

Answer the following questions if you answer "**Yes**" to any of the above questions.
If the answer is "**Yes**" to **Q1**, please answer the following questions:

		Yes	No
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		

If the answer is "**Yes**" to **Q2**, please answer the following questions:

		Yes	No
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)	X	
Q8	Are the third-party data used from an open source?	X	
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is "**Yes**", do you have the approval to use third-party data sets? Attach the proof to PSQ application.		

If the answer is "**Yes**" to Q5, please answer the following questions:

		Yes	No
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		
Q12	Will there be a chance for any conflict of interest between the parties? [If Yes, provide details of the issue and your plan to solve it]		